

Information-Geometrical Perspectives of Regime Switching in Stock Markets

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Abstract

In a financial market, regime switching occurs as a result of a financial stress causing volatility to change rapidly. In this article, we present an information geometry framework for analyzing and examining market behaviors pertaining to regime switching. We demonstrate our method on stock prices of the Standard & Poor's 500 index (S&P 500) from 1995 to 2022, and the STOXX Europe 600 index (STOXX 600) from 2004 to 2022, which endured multiple crises. Fisher information distance (FID) and angle (FIA) are shown as potent tools for tracking high movement in the distribution of financial returns and clustering groups of statistical parameters. We also compare the results of regime switching from FID and FIA to those from the Hidden Markov model (HMM) to demonstrate that FID and FIA can be effective tools for capturing the level of regime switching. These findings contribute to the understanding of financial regime switching from the perspectives of information geometry, which probe the deformation distribution of fundamental level rather than the statistical moments at manifestation level.

Keywords: financial return distribution, regime switching, information geometric measurement, Fisher information, hidden Markov model

1. Introduction

Regime switching is a phenomenon when a system changes its characteristics abruptly. It is commonly found in complex systems, for instance, in economic systems as financial markets undergo business cycles or bubbles or crashes of financial asset prices. A regime switching in a financial market can emerge from changes in fiscal and monetary policy, economic factors, fundamentals of underlying financial assets, or even from internal dynamics of agents without external influence [1–3].

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A simple model of a regime switching is a Markov switching (MS) or hidden Markov model (HMM), in which the switching between two regimes is modelled by a Markov process. Goldfeld and Quandt first introduced this model in 1973 [4] to describe a common pattern of time series in economics. Later, in 1989, Hamilton provided an algorithm to determine the probability of the Markov switching from real data for the autoregressive model [5]. Kim generalized the regime-switching model for a state-space model, making it more applicable to a broader range of applications [6].

For financial markets, various regime switching models had been investigated in different types of markets, including stock markets and foreign exchange markets [7–9], interest rates [10–16] and risk of bond yields [17]. Ang and Chen showed that a regime-switching model could capture asymmetric correlation in the stock price data more effectively than the multivariate normal distribution [18]. Different behaviors in interest rate often correspond to the changes in monetary policy, inflation rate, or business cycle. For foreign exchange rates, Engel and Hamilton examined the U.S. dollar and found that the regime-switching model could describe the data better [19]. Bekaert and Hodrick also demonstrated regime shifts in many currencies [20].

The aforementioned literature reports suggest the importance of regime-switching in financial markets. However, the drawbacks of existing regime-switching models stem from they allow only a few regimes for describing the time series, and so the functional structure of a model is pre-determined. Subsequently, we propose an alternative method based on information geometry to probe the regime change of stock markets. This method utilizes tools from differential geometry to determine a distance between two probability distributions in a market from consecutive time windows and then uses such distance to describe the dynamics and regime switching of the market instead. Abundant mathematical tools from differential geometry allow us to investigate the phenomena from a new perspective thoroughly [21–23]. In our point of view, information contained in the data distribution is more fundamental than the manifested mean, variance, or any statistical moment. To that end, we aim to probe the changes in the market behaviors directly from examining the distribution of returns rather than its volatility.

Since originated from the work of Amari [21, 24] and culminated in his book [25], information geometry has play impacting roles information science, especially in finance and medicine [26–28]. Fisher information has been used to define a metric on a Riemannian manifold of probability distributions, which is invariant under reparametrization of coordinates. Each point in this statistical manifold corresponds to a probability distribution. Thus, it provides a geometrical visualization and measure of closeness, which can be viewed in terms of dynamical change. With these desirable characteristics, many applications of information geometry have been realized in various fields, such as music information processing [29], neural networks [30], image processing [31–33], statistical physics [34–38], sensing systems [39–41], machine learning [42–44], mathematical biology [45], and coding [46]. In the field of finance, Brody and Hughston transformed term structures of interest rates into probability densities, from which they used information geometry to analyze the relationship between yield curves and reformulated the theory of interest rate dynamics, which gave rise to a new method of interest rate modeling [47–49]. Similarly, Gao et al. defined a manifold of

an equivalent martingale measure in the sense of information geometry and derived asset-pricing technique in an incomplete market [50]. Liang and Sun used information geometry for machine learning algorithm, which resulted in improvement of stock price prediction [51]. Moreover, information geometry also has been applied to risk measurement of financial assets by defining a more accurate relation between the asset prices [52], or making the link between micro- and macro-economics and examining their evolving correlation [53].

Fisher information distance (FID) and Fisher information angle (FIA) have recently been used to capture changes in many financial models [54–56]. For instance, financial network dynamics can indicate evolutionary systemic risks, and FID can be used together with the minimal spanning tree method to construct the inherent structured network of stocks, from which the average FID provides the evolution index of the stock’s network for comparison in various financial situations [54]. It can also be applied to categorize groups of stocks through hierarchical clustering [55]. Furthermore, FID yields robust clusters than the correlation distance when compared to the entropy heat map of stocks [56].

We observed that, in literature, FID between stocks is commonly examined for a given time period with constant time window. Hence, in this study, we investigate an alternative point of view by calculating FID between the return distributions of a stock at different times. We show how changes in the distributions can be interpreted as regime switching, albeit one with continuous states. More importantly, we compare the performance of a regime switching model via FID to a typical model like HMM. Essentially, we propose a new method for analyzing regime switching in financial asset prices, based on information geometry. It stipulates that as the financial asset returns can be expressed in a probability distribution, whose change in the structures of distribution reflects the potential regime change, information geometry appears to be an appropriate and powerful method for such change, and it can provide more insights of regime switching in financial markets.

This article is organized into five sections. Section 2 provides theoretical foundation and outline of analysis of regime switching using information geometry and HMM. In Section 3, we present results and discussion of the Fisher information metrics in the Standard & Poor’s 500 index (S&P 500) when compared to HMM. Finally, we summarize the findings of this research and remark on their application in Section 4. The same analysis for STOXX Europe 600 index (STOXX 600) is provided in the appendix.

2. Methodology

2.1. Data

We use the daily price ($p(t)$) of S&P 500 from 1995 to 2022. This index comprises of stocks of 500 large capitalization companies, which are traded in U.S. equity markets, weighted by their market capitalization. It is one of the most common indices representing U.S. equity markets as a whole, which is one of the largest and most developed markets in the world. The chosen time duration covers many crises that caused prominent regime switching, i.e., Russian default in 1998, dot-com crash in early 2000s, credit crunch in 2008, and COVID pandemic effects in 2020. These data need an adjustment of corporate actions,

such as dividend payment and par splitting. However, we do not directly deal with the index time series. Instead, we analyze the logarithmic daily return of the index defined by

$$r_\tau(t) = \log p(t + \tau) - \log p(t) \quad (1)$$

To further test the method robustness, we perform the same analysis for STOXX 600 from 2004 to 2022, whose results are presented in Appendix D.

2.2. Software Tools

In this work, we use MATLAB version R2022a to analyze data. MATLAB provides potent toolboxes for analysis and methods in financial markets, such as econometrics toolbox and curve fitting toolbox [57]. To examine regimes in financial markets with HMM, we utilize the *hmmlearn* library in Python [58].

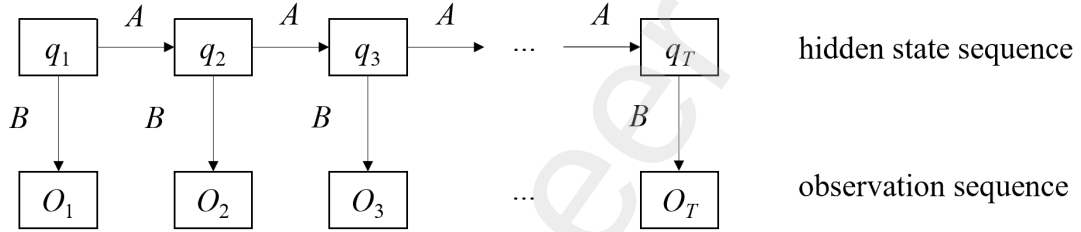


Fig. 1: Visualization of a HMM process. This diagram is adapted from Ref.[59].

2.3. Hidden Markov Model (HMM)

A hidden Markov model is a probabilistic model which uses observation sequence to identify hidden sequence based on a Markov process. The model is composed of three model parameters $\lambda = \{A, B, \pi\}$, where A is the state transition probabilities; B is the observation probability matrix (emission probabilities); and π is the initial state distribution. Fig.1 illustrates a HMM model with the observation length T . We can compute the probability of the state sequence $P(Q, O)$ given an initial state π . Suppose we consider the hidden state having two states $Q = \{H, C\}$ and the observation sequence having three observation states $O = \{S, M, L\}$, then the transition probability and the emission probability are a matrix with a size 2×2 and 2×3 , respectively. For instance, by known observation sequence $SMSL$, the HMM model will find the maximum probability of $P(Q, O)$. In other words, the model evaluates sequences of the hidden state that can generate $SMSL$. These sequences can come in various configurations of states, such as $HHHH$, $HHHC$, or $CCCC$. The sequence that gives the highest probability will be selected to model the observation.

The preceding example is merely for demonstration purposes. We note that an observation can be a continuous state generated from an underlying probability distribution. Just as many researchers have developed HMMs to analyze the dynamics of market states [60–62], in this research, we use returns as an observation sequence to identify hidden market

states, namely those of low volatility and high volatility. To that end, we employ the Viterbi algorithm, the Forward-Backward algorithm, and the Baum-Welch algorithm in the parts of the optimization and estimation of HMM. For more details of these algorithms, please see Refs. [58, 59, 63, 64].

2.4. Time Series Analysis with Information Geometry

First, at a given time horizon of daily data, we need to construct the probability distribution of the logarithmic returns during a time window of a given length. Then, the time window is moved along the time series by its daily trading data, to construct a new probability distribution of the logarithmic returns. Repeating the process for the whole time series, we have created a set of probability distributions that exhibit changes over time, which can represent the regime switching by the abrupt change of their profiles. To analyze the change of probability distributions in time, we employ the Fisher information metric [65].

$$g_{ij}(\theta) = \sum_{i,j} \left(\frac{\partial}{\partial \theta_i} \log p(x|\theta) \right) \left(\frac{\partial}{\partial \theta_j} \log p(x|\theta) \right) p(x|\theta), \quad (2)$$

where $p(x|\theta)$ is a probability distribution function of random variable x with a set of parameters θ . Here, θ play a role of coordinates on a manifold. In this case, the random variable x represents the logarithmic returns of the index. It should be noted that, the Fisher information metric is an infinitesimal version of the Kullback-Leibler divergence or the so-called relative entropy. Now, with this metric, we can define an inner product $\langle D, D' \rangle_p$ and the length $\|D\|$ of tangent vectors D and D' at the point p on the manifold by [25]

$$\langle D, D' \rangle_p = g_{ij}(\theta_p) D^i D'^j \quad (3)$$

and

$$\|D\| = \langle D, D \rangle_p = g_{ij}(\theta_p) D^i D^j, \quad (4)$$

respectively. Here, the Einstein summation notation is used. Given an arbitrary curve $\theta(t)$, which is parameterized by t linking two points a and b on the manifold, the length of this curve can be defined as

$$\|\theta\| = \int_a^b \left\| \frac{d\theta}{dt} \right\| dt = \int_a^b \sqrt{g_{ij} \dot{\theta}^i \dot{\theta}^j} dt, \quad (5)$$

where $\dot{\theta}^i$ is the first order derivative of θ^i with respect to t . The distance on a Riemannian manifold is defined as the shortest path linking two points. The equation above is minimized via the calculus of variation, and we get the Euler-Lagrange equations [65]

$$g_{ki} \ddot{\theta}^i + \Gamma_{k,ij} \dot{\theta}^i \dot{\theta}^j = 0, \quad (6)$$

where $\ddot{\theta}^i$ is the second order derivative of θ^i with respect to t , and $\Gamma_{k,ij}$ is the Christoffel symbol

$$\Gamma_{k,ij} = \frac{1}{2} \left\{ \frac{\partial g_{ik}}{\partial \theta^j} + \frac{\partial g_{kj}}{\partial \theta^i} - \frac{\partial g_{ij}}{\partial \theta^k} \right\}. \quad (7)$$

The solution of the Euler-Lagrange above is a geodesic or shortest path, and its length is the distance between two probability distributions that we would like to measure.

2.5. Fisher Metrics of Normal Distribution

The return distribution in a financial market typically exhibits a leptokurtic distribution with kurtosis greater than three. As one of the stylized facts in financial markets [66, 67], this demonstrates a heavy tail distribution. However, we choose a normal approximation of the return distribution because of its simplicity and derivable analytical expression, hence less time-consuming in computation. More importantly, it still remains efficient in capturing the profile of return distribution, as indicated by its mean and standard deviation (SD). Readily, FID between two normal distributions $N(\mu_1, \sigma_1)$ and $N(\mu_2, \sigma_2)$ can be obtained analytically as [22]

$$D_F((\mu_1, \sigma_1), (\mu_2, \sigma_2)) = \sqrt{2} \ln \left(\frac{F + (\mu_1 - \mu_2)^2 + 2(\sigma_1^2 + \sigma_2^2)}{4\sigma_1\sigma_2} \right) \quad (8)$$

where

$$F = \sqrt{((\mu_1 - \mu_2)^2 + 2(\sigma_1 - \sigma_2)^2)((\mu_1 - \mu_2)^2 + 2(\sigma_1 + \sigma_2)^2)}. \quad (9)$$

Additionally, we define the angle of a vector created by two points on the manifold and a vector that is parallel to the x axis (μ) as FIA, which can measure the direction of changing regime. FIA is considered only in a range of 0 to 2π , referring to a change in $\Delta\mu$ and $\Delta\sigma$. For example, $\theta \in [3\pi/2, 2\pi]$ indicates that $\Delta\mu$ is increasing, whereas $\Delta\sigma$ is decreasing. The detail of derivation of FIA is included in Appendix C. We can see that FIA between two normal distributions is given by

$$\theta = \begin{cases} \cos^{-1} \left(\frac{\mu_2 - \mu_1}{\sqrt{(\mu_2 - \mu_1)^2 + 2(\sigma_2 - \sigma_1)^2}} \right), & \sigma_2 \geq \sigma_1, \\ -\cos^{-1} \left(\frac{\mu_2 - \mu_1}{\sqrt{(\mu_2 - \mu_1)^2 + 2(\sigma_2 - \sigma_1)^2}} \right), & \sigma_2 \leq \sigma_1. \end{cases} \quad (10)$$

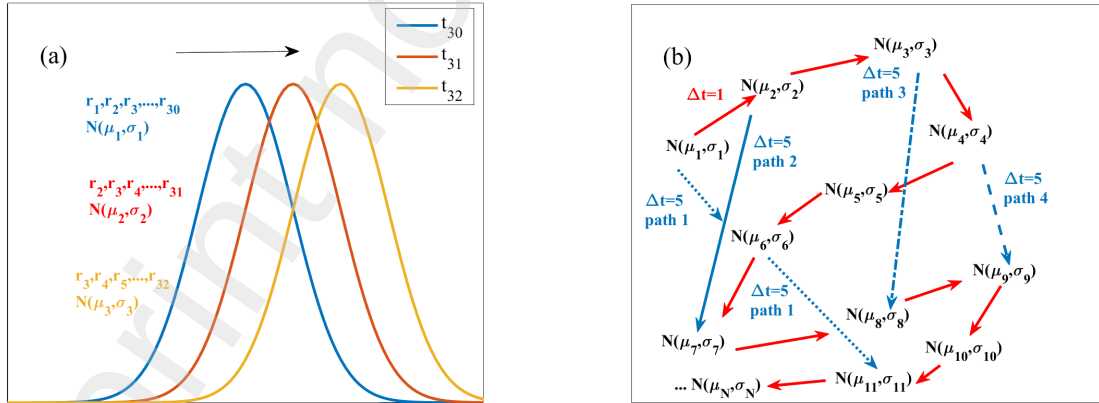


Fig. 2: Illustrations how to construct a return distribution at time t and regime switching of the return distributions with varying lag times Δt : (a) moving daily return distribution by using data from 30 trading days prior; (b) an example of computation of the Fisher information metric between t and $t + \Delta t$ with different lag times. Various paths of regime switching are considered at a certain lag time. For instance, $\Delta t = 5$ has different routes depending on return distribution at time t as shown in various blue paths.

2.6. Analysis Procedure

The regime switching in S&P 500 will be analyzed using these information geometric metrics, and will be compared to the results from HMM. Our next step is selecting a suitable length of the time window used to construct a single distribution of the returns. We perform moving daily return distribution using collected data from the prior 30 trading days, as shown in Fig. 2(a). As an example, paths on statistical manifold between distributions in varied Δt are depicted in Fig. 2(b).

After determining moving daily return distributions, the daily FID at time t is calculated as the distance between two daily return distributions at time t and $t - 1$ by means of the Fisher information metric. A date which the return distribution changes abruptly indicates that the market switches regimes along with considerable change in market conditions. We explore both the FID and FIA signals with varying lag times Δt to see tail behaviors change as the lag time increases, and also to compare performance of classifying clusters of statistical parameters between HMM and FID.

3. Results and Discussion

3.1. Data and Return Distributions

We begin our investigation by computing the daily logarithmic returns (as defined by equation (1)) of the S&P 500 from logarithmic prices from 1995 to 2022, as shown in Figs.3(a-b). The first and second statistical moments of the daily logarithmic return time series with various moving window sizes of 10, 30, 60, 120, 180, and 240 trading days are presented in Fig.3(c) in the scale normalized by the averaged SD. Moreover, we examine a thousand of shuffled return time series by using random permutation, from which we can assess if detected features are significant. This will be demonstrated in the following section. As shown in Fig.3(d), we have chosen a time window size of 30 trading days for our investigation since would like to approximate the return distribution by the normal distribution for the sake of simplicity in FID calculation. The kurtosis of distribution in this time window is approximately to three, indicating a normal distribution is reasonable. A kurtosis value less than three denotes a lower peak distribution known as platykurtic, and that greater than three implies a higher peak distribution known as leptokurtic when compared with a normal distribution. As such, the tail of the platykurtic distribution is thicker than that of the leptokurtic one. We also test the normality of the distribution from each size of the time window, as shown in Appendix A. Interestingly, larger window sizes increase the likelihood of time windows wherein data distributions are not normal.

Fig.3(c)-(d) depict statistical moments as the time window size varies. The mean and variance of moving return time series are nearly zero in all cases. Yet, they display negative skewness as the size of the time window increases, suggesting that expanding time windows size induces asymmetry of the return distribution with a longer left tail. Furthermore, kurtosis increases as the size of the time window widens, implying that the distribution becomes more leptokurtic. As a consequence, we select 30 trading days as the time frame to evaluate since the return time series size is appropriately not too small for approximating the return distribution with the normal distribution. We also remark that the skewness

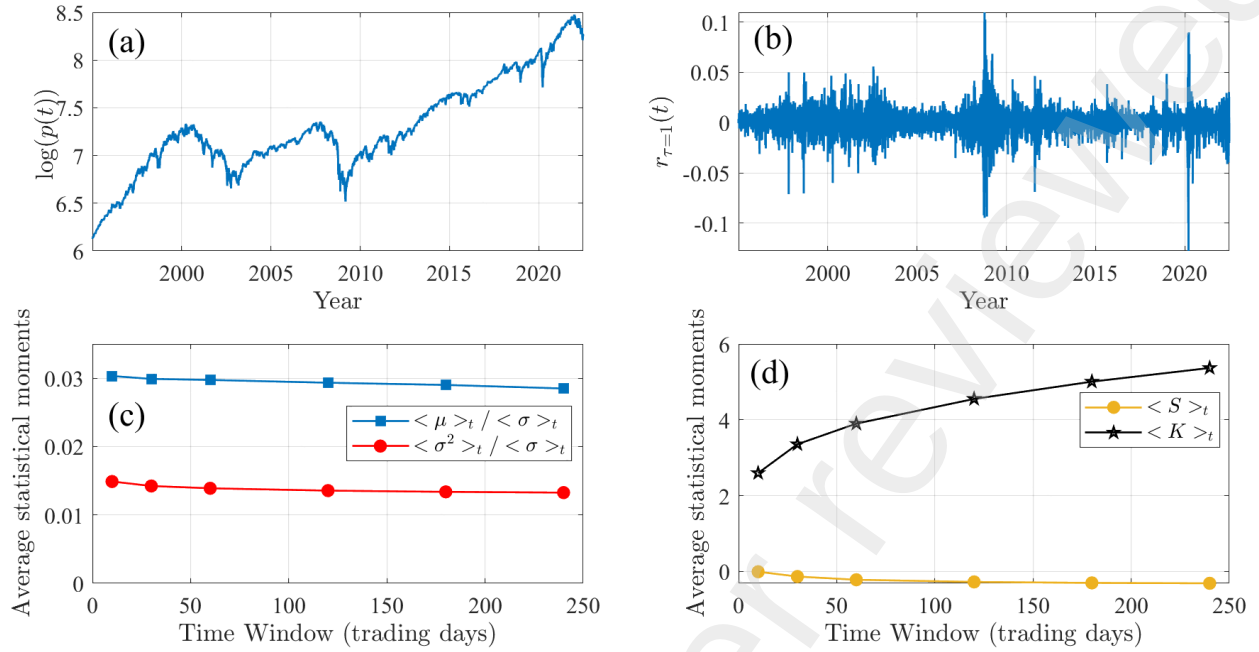


Fig. 3: Financial data and statistical moments: (a) the logarithmic price of S&P 500 from 1995 to 2022; (b) the daily logarithmic return of S&P 500 from 1995 to 2022; (c) the average mean and variance in the standard deviation unit of the daily logarithmic return distributions with different sizes of time windows; (d) the average skewness ($\langle S \rangle_t$) and kurtosis ($\langle K \rangle_t$) of daily logarithmic return distribution with different sizes of time windows.

become more negative as the time window expands because as the time window grows, more data is included in the time series, which may be an effect of the financial crisis, causing statistical metrics to abruptly alter in a negative way. As a result, the negative tail of the return distribution is prolonged. In contrast, the kurtosis increases as the time window stretches. This is due to the fact that most return values are approaching zero. Additionally, the increased number of data for the distribution is obviously resulting in this stylized fact, leading to higher kurtosis. As such, there is low possibility of obtaining an extreme return value; however, these extreme values would be higher during a financial crisis, causing the shape of distribution to become more leptokurtic.

3.2. Distributions of FID and FIA

In principle, the change in a market structure is manifested in the statistical moments, but fundamentally it should be expressed in the deformation of the return distribution, which can be viewed from an information geometric aspect. In this section, we aim to provide evidence that regime switching can be reflected in the FID or FIA characteristics. To visualize FID and FIA, we make scatter plots to show the relation between the change in μ and σ as shown in Figs.4 and 5. It can be seen that FID depends on a level of change in mean and standard deviation of the return distribution. A lower FID (dark blue) corresponds to lower changes, while a larger FID (yellow) indicates a higher change in both μ and σ .

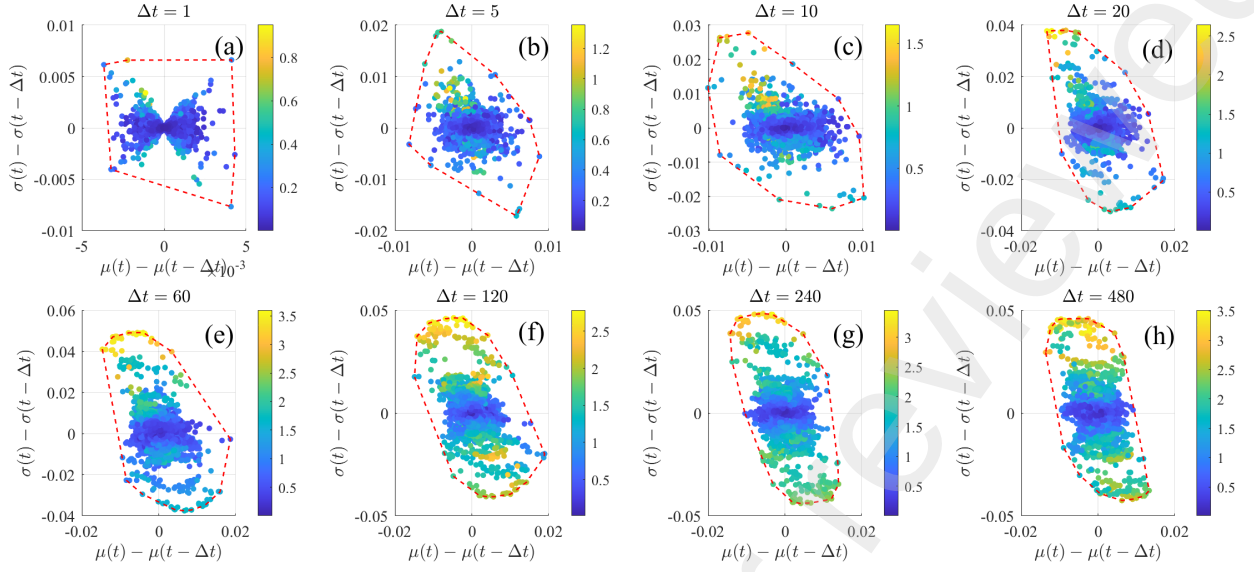


Fig. 4: Scatter plots of changes in μ and σ : (a) $\Delta t = 1$; (b) $\Delta t = 5$; (c) $\Delta t = 10$; (d) $\Delta t = 20$; (e) $\Delta t = 60$; (f) $\Delta t = 120$; (g) $\Delta t = 240$; (h) $\Delta t = 480$. The color of the scatters indicates the strength of FID, whereas the red dashed line denotes a convex hull region of scattering.

222 An absolute large FIA indicates a regime of negative mean change ($\mu(t) - \mu(t - \Delta t) < 0$),
 223 while an absolute low FIA shows a regime of positive mean change ($\mu(t) - \mu(t - \Delta t) > 0$).
 224 In both cases, the standard deviation change is close to zero ($\sigma(t) - \sigma(t - \Delta t) \approx 0$).

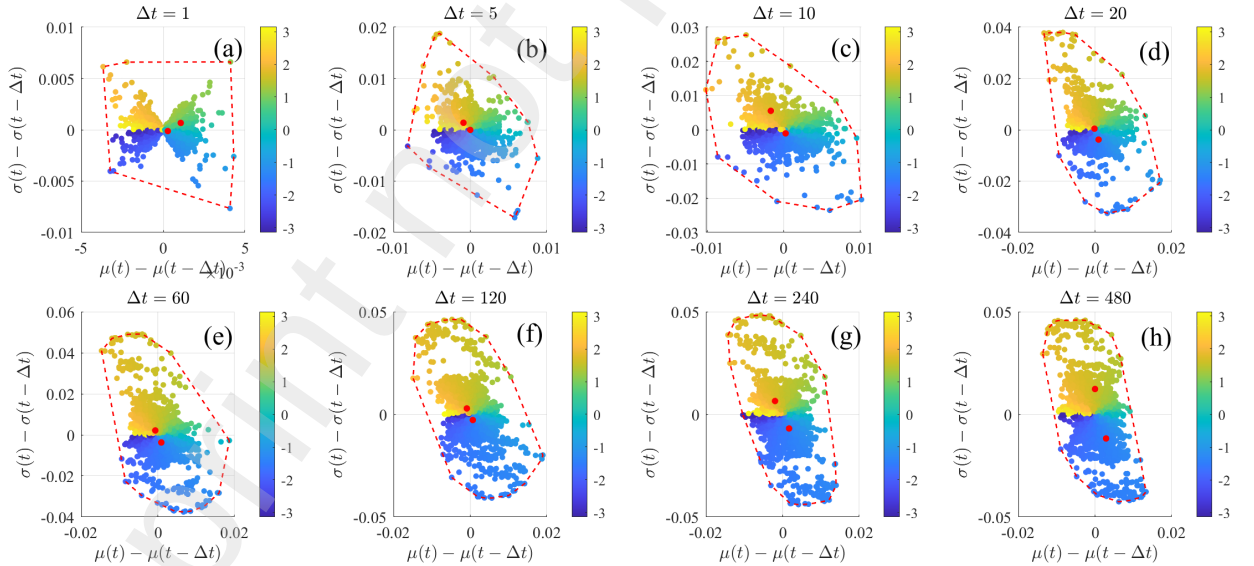


Fig. 5: Scatter plot of changes in μ and σ : (a) $\Delta t = 1$; (b) $\Delta t = 5$; (c) $\Delta t = 10$; (d) $\Delta t = 20$; (e) $\Delta t = 60$; (f) $\Delta t = 120$; (g) $\Delta t = 240$; (h) $\Delta t = 480$. The color of the scatter indicates the strength of FIA, whereas the red dashed line denotes a convex hull region of scattering. The red dots depict the regions where positive and negative Fisher information angles are most likely to be found.

We present the probability density functions of FID and FIA compared to the data from shuffled return time series to examine significant features in the empirical return distribution. For FID, the unshuffled data shows long right tail distribution which gets longer as the lag time increases, as shown in Fig.6. In the context of regime switching, this observation highlights the occurrence of infrequent but substantial jumps or extreme changes in the return distribution within a financial market. Having compared to the shuffled data at smaller lag time, the tail of the FID distribution of untampered data is shorter than that of the shuffled data, and the shuffled one gets shorter as the lag time increases. We note that FID provides the opposite trend distribution when compared to stylized facts of the return distribution. The cumulative distribution function of the absolute returns tend to have progressively shorter tail and approaches the Gaussian distribution as the lag time increases. FID, on the other hand, exhibits a longer tail as the lag time increases. When compared to a smaller lag time, a greater lag time makes a significant difference due to a commonly observed volatility clustering phenomena. As a result, they have the opposite tail trends. From one perspective, the FID may be viewed as a shift in the return distribution.

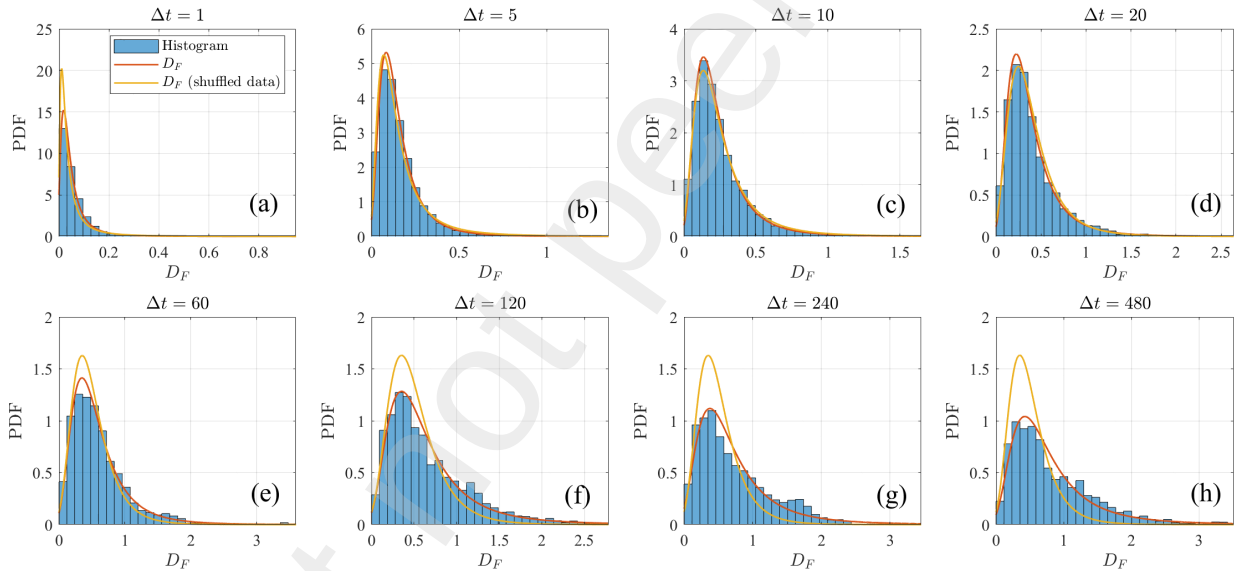


Fig. 6: The probability density functions (PDF) of FID from original and shuffled data of returns: (a) $\Delta t = 1$; (b) $\Delta t = 5$; (c) $\Delta t = 10$; (d) $\Delta t = 20$; (e) $\Delta t = 60$; (f) $\Delta t = 120$; (g) $\Delta t = 240$; (h) $\Delta t = 480$ days. Red lines are the fitting of real data generalized extreme value distributions, whereas the yellow lines depict the average of the FID distributions from the shuffled data 1000 configurations.

The FIA distribution exhibits three peaks at $\Delta t = 1$ day; however, it becomes two peaks as the lag time increases with disparing asymmetry between the two peaks. The distribution of real data and that of shuffled data are very similar for a small lag time, while the peak of shuffled data is lower than that of the real data for a large lag time as shown in Fig.7. Hence, ordering of data affects the changes in the Fisher information metrics as it yields different distributions. In Figs.5 and 7, the red dots represent the most likely positive and negative Fisher information angles. When compared between different scatter plots, it can

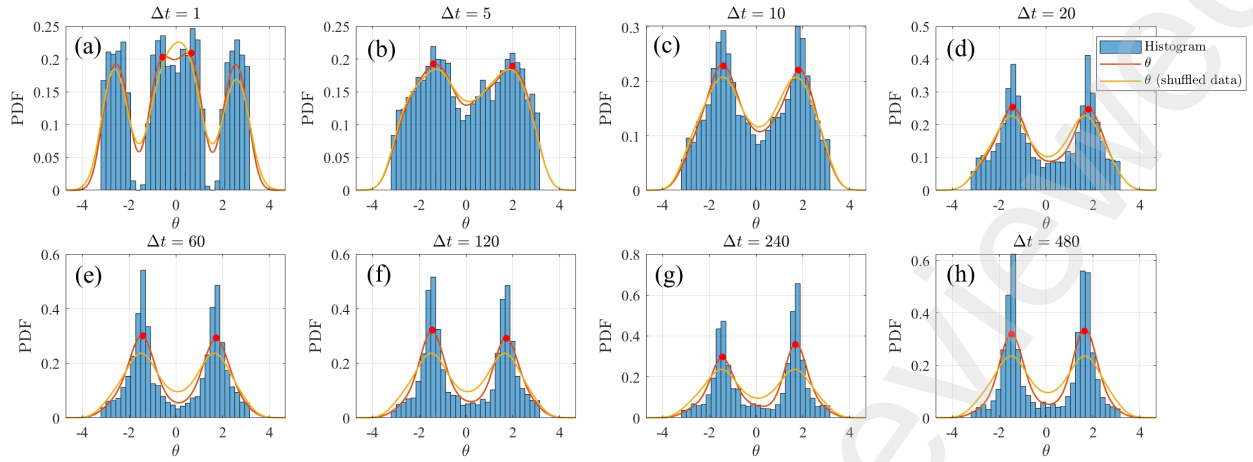


Fig. 7: The probability density function (PDF) of FIA from original and shuffled data of returns: (a) $\Delta t = 1$; (b) $\Delta t = 5$; (c) $\Delta t = 10$; (d) $\Delta t = 20$; (e) $\Delta t = 60$; (f) $\Delta t = 120$; (g) $\Delta t = 240$; (h) $\Delta t = 480$ days. Red lines are the fitting of real data Kernel distributions, whereas the yellow lines depict the average of the Fisher information angle distributions from shuffled data 1000 configurations.

be observed that the most likely negative and positive FIA are relatively closer to each other for a small lag time, and drift further apart as the lag time increases, for $\Delta t \geq 20$ days. From information geometry, the drifting of FID and FIA could signify a structural change of a financial market and should be investigated for regime switching.

It is evident from Fig.7 that the distributions coming from the real data are more asymmetric than that from the shuffled data. This is reasonable because, in a financial market, the direction of changing regime measured by FIA depends on the financial situation at that specific moment, whereas the shuffled data of returns represent random order of time series. Consequently, the distributions of FIA from the real data and shuffled data are significantly different.

The real data yields higher peaks in the distribution of the FIA than does the shuffled data. This may be due to the nature of evolving statistical moments (μ and σ) in financial markets, which often incur leptokurtic distributions for financial quantities. For instance, the returns normally fluctuate around zero with high probability, but in some financial situations or crises, the returns deviate to higher values. When we change the arrangement of the return time series by shuffling, the characteristics of statistical moments also change, showing the peaks of the shuffled data lower than those of the real data.

The relation between the changes in the mean and standard deviation is also examined. It is apparent that increasing Δt brings more scattering in values and yields higher negative correlation. We compute a convex hull area to measure the size of scattering as shown in Fig.8(a) and correlation between the changes in the mean μ and standard deviation σ as shown in Fig.8(b). Convex hulls are often used to visualize a set of two-dimensional sample points to compare the dispersion of extreme data. In our case, as the lag time increases, the convex hull area increases, implying that the extreme values ($\Delta\mu$ and $\Delta\sigma$) disperse more. This can be explained by the fact that the financial quantities are influenced

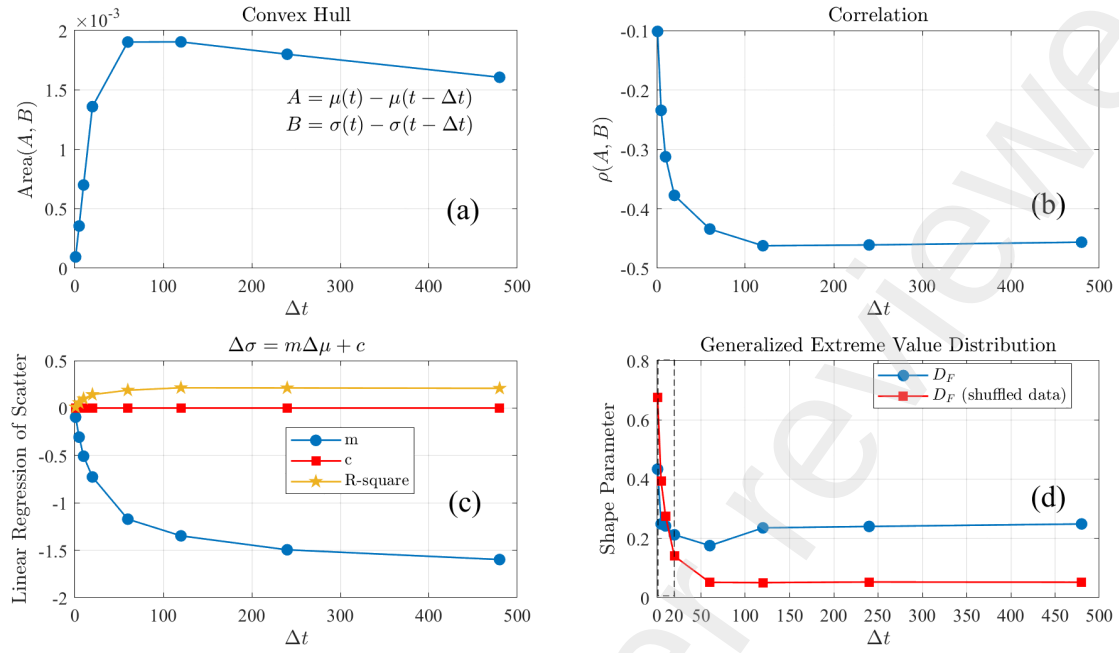


Fig. 8: Statistical features of change in mean μ and standard deviation σ : (a) the convex hull area of scattering region; (b) correlation between changes in μ and σ ; (c) linear regression of scatter; (d) comparison of tail index between real data and shuffled data with the generalized extreme value distribution.

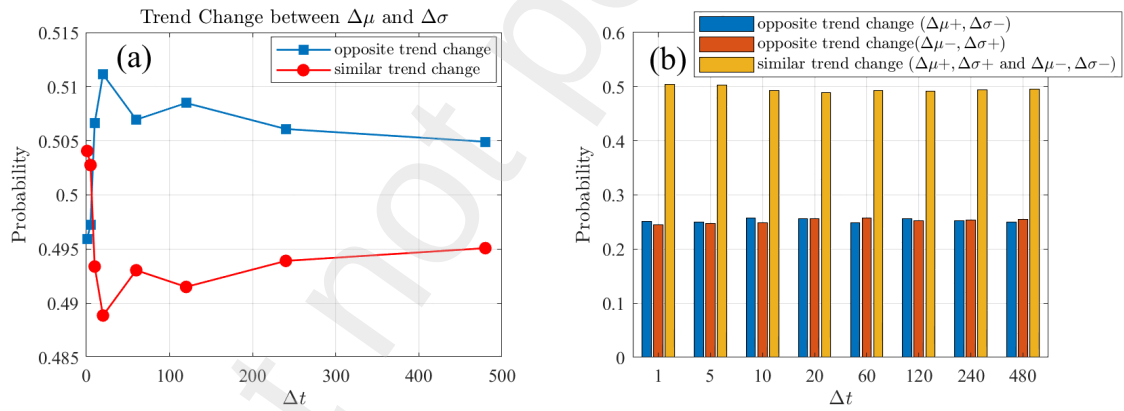


Fig. 9: Comparison of trend changes between $\Delta\mu$ and $\Delta\sigma$ as the lag time Δt increases: (a) the probability of trend changes between $\Delta\mu$ and $\Delta\sigma$; (b) decomposition of the probability of opposite trend changes between $\Delta\mu$ and $\Delta\sigma$. The plus symbol represents an increased value in the signal; the minus symbol a decreased value. In (b), the dark blue and orange bars indicate opposite trend changes between signals, and the yellow ones a similar trend change.

by different financial circumstances; when measured at far-away points in time, they tend to show different behaviors. Similarly, when the lag time increases, the changes in μ and σ become more anticorrelated, implying that as $\Delta\mu$ increases, $\Delta\sigma$ tends to decrease, and vice versa. Furthermore, as shown in Fig.8(c), the linear regression between $\Delta\sigma$ and $\Delta\mu$ confirms the negative-slope between them. When the lag time Δt increases, the slope of the

linear regression becomes more negative but saturated. As $\Delta\mu$ and $\Delta\sigma$ show anticorrelation, one wonders which scenario is more likely: (i) $\Delta\mu$ increases and $\Delta\sigma$ decreases or (ii), $\Delta\mu$ decreases and $\Delta\sigma$ increases. It turns out both scenarios are equally likely in S&P 500, and together comprising slightly more than 50% of events when $\Delta t \geq 10$ days, as shown in Fig.9(a) and Fig.9(b). This is reasonable, especially during a financial crisis when a stock market is volatile, and most traders get a negative averaged financial return. Moreover, we fit FID with the generalized extreme value distribution, which commonly is used to model the smallest values or largest values of observed data [55, 68], to examine the distribution tail as the lag time changes. The results are presented in Fig.8(d). The shape index of FID from the shuffled data (red square) is greater than that of the real data at the beginning, but the trend changes after $\Delta t = 20$ days when the real data have a greater value in the shape index, suggesting that it has a longer tail. This is influenced by the selected time window as mentioned in Section 3.1, which indicates that expanding the time window considerably changes the statistical moments of the return distribution. Similarly, because of the different structure of the return distribution, a longer lag time tends to yield higher shape index of FID.

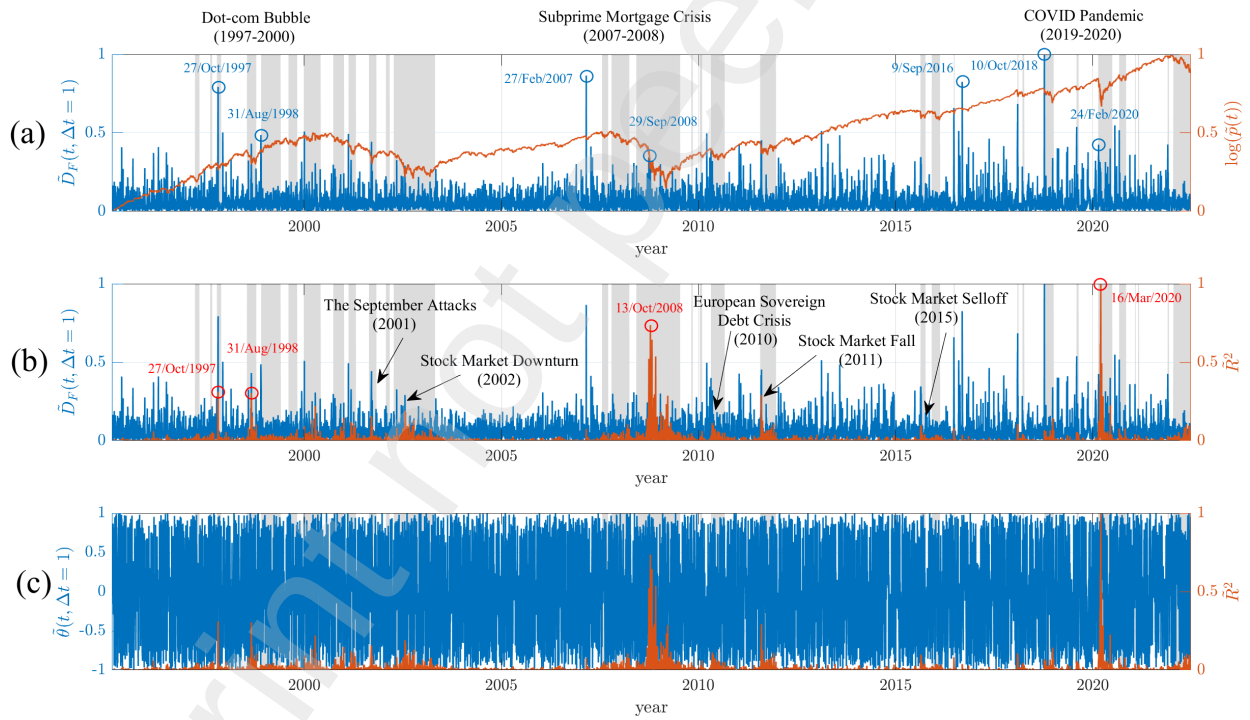


Fig. 10: Superposed graphs of normalized FID and financial quantities in different financial periods of S&P 500: (a) normalized FID versus normalized logarithmic price; (b) normalized FID versus normalized volatility $\tilde{R}^2$; (c) normalized FIA versus $\tilde{R}^2$. The gray-shaded areas illustrate high volatility periods defined by HMM; the black arrows indicate short periods of high volatility. The blue circles represent notable peaks of normalized FID which occurred on specified dates.

3.3. Time Series of Fisher Information Metrics and Regime Switching

To examine the Fisher information time series, we plot the daily FID ($\Delta t = 1$ day) in superposed graphs with the logarithmic price and volatility computed by the daily logarithmic returns squared ($R(t)^2 := r_{\tau=1}(t)^2$), as shown in Figs.10(a-b), respectively, to see how the FID changes its behavior during financial crises. To compare signals within the same unit, we normalize all quantities to the range of $[0, 1]$ for FID, $\log(p(t))$, and R^2 by

$$\tilde{X} = \frac{X - X_{min}}{X_{max} - X_{min}}. \quad (11)$$

The normalized FIA time series with $\Delta t = 1$ day is also shown in Fig.10(c). A higher lag time causes a delay of a distinct peak during the period of a financial crisis because the return data in the distribution does not include the dates with high volatility. As a consequence, selecting $\Delta t = 1$ day should yield more precise information to capture the dates when the volatility changes significantly. As expected, large changes in volatility occurred amid several periods of financial crises, and also coincided in grey regions indicating regime switching by HMM. Evidently, normalized FID can capture large changes in the volatility, yielding the maximum peak on observedly the same date. For instance, both the normalized FID and $\tilde{R}^2$ reached their peaks on October 27, 1997 and August 31, 1998 during the dot-com bubble period. During the period of subprime mortgage crisis, the highest volatility occurred on October 13, 2008. FID is still high, but not as high as it was on February 27, 2007. The nearest distinct level of normalized FID occurred on September 29, 2008, indicating that there is a large deformation of the return distribution. Note that this was just after the bankruptcy of Lehman Brothers on September 15, 2008. It also suggests that normalized FID can provide additional information because it shows a distinct peak even if the date of high volatility $\tilde{R}^2$ is not present. On that date, there is a significant change in the return distribution compared to that on the previous date as it shows the distinct normalized FID. Furthermore, FID can provide level of regime switching when HMM cannot, such as between 2010 and 2016. During the COVID pandemic period, there is a peak on February 24, 2020 before $\tilde{R}^2$ reached its peak on March 16, 2020. This deformation of the return distributions reflected in FID suggests a high level of regime switching.

In contrast, FIA appears to provide less significant features than FID because of its presistent fluctuation over time. As a result, we are unable to obtain relevant concrete information from FIA to detect significant changes in volatility or a financial crisis. In principle, FID can quantify the level of regime switching (amplitude) according to the changes in the return distributions, whereas FIA can quantify the direction of change. Unlike other traditional indicators that map a shift in the return distributions to two regimes (say high volatility regime and low volatility regime), our investigation utilizes information geometry to provide infinite regime switching on the return distribution. Consequently, it includes more precise information about the continuous change of the return distribution over time in various financial situations.

There can be several significant peaks over a crisis period, as exhibited in Fig.10. We find that FID generally shows a distinct peak when R^2 is high. However, it does not necessarily

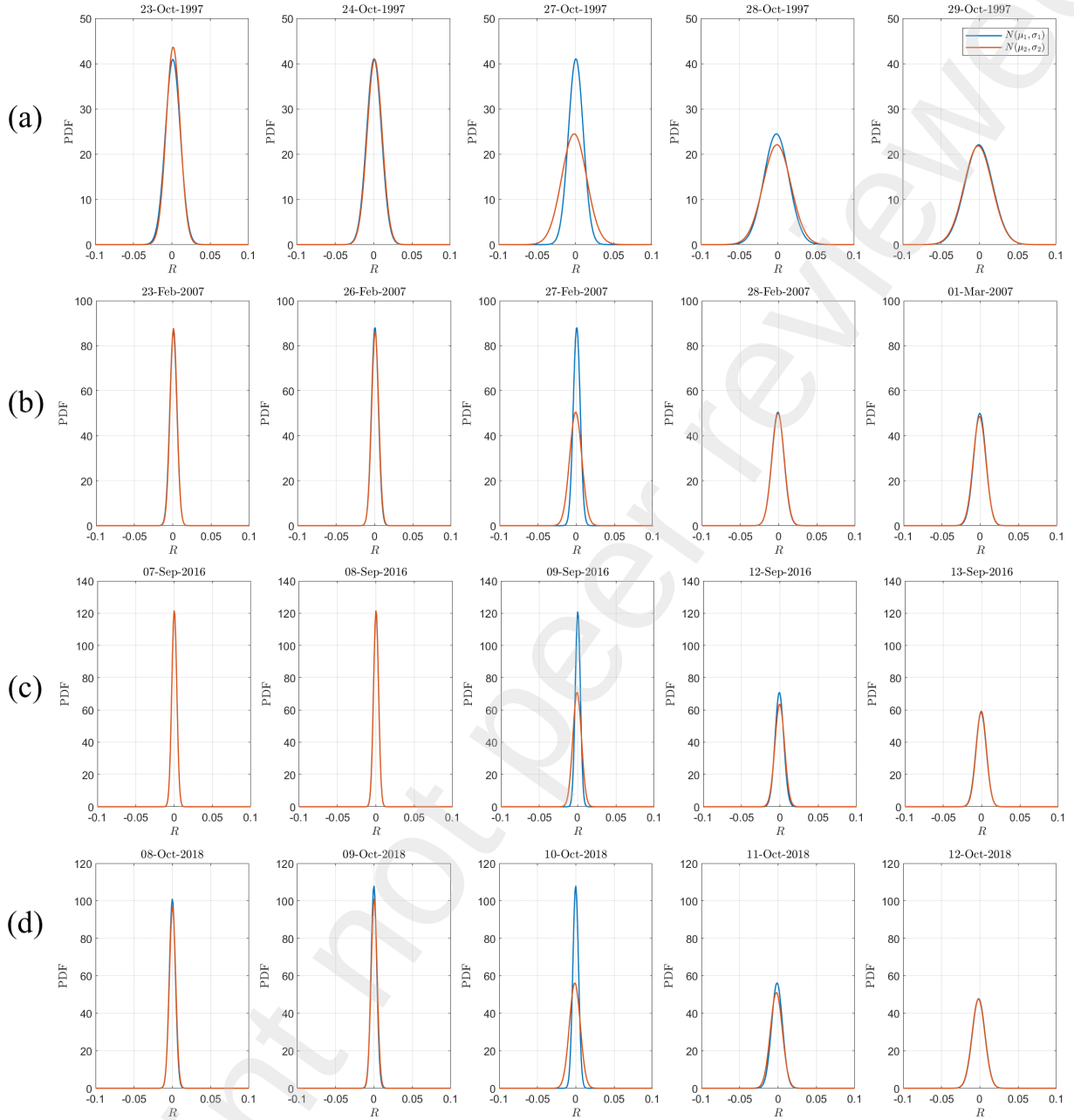


Fig. 11: Graphs exhibit the return distributions in the S&P 500 market that correspond to FID peaks in several periods: (a) 1997, (b) 2007, (c) 2016, and (d) 2018. Top four highest FID peaks are chosen.

depend on R^2 only. As noted, a high value of FID represents the level of changing in the return distribution. Therefore, the peak in FID might not appear together with a high value of R^2 , as FID can have high peaks even when R^2 is low, see Appendix B for detailed analysis. More evidently, the visualization of the return distributions before and after the peak is illustrated in Fig.11. It is apparent that the shape of the return distribution

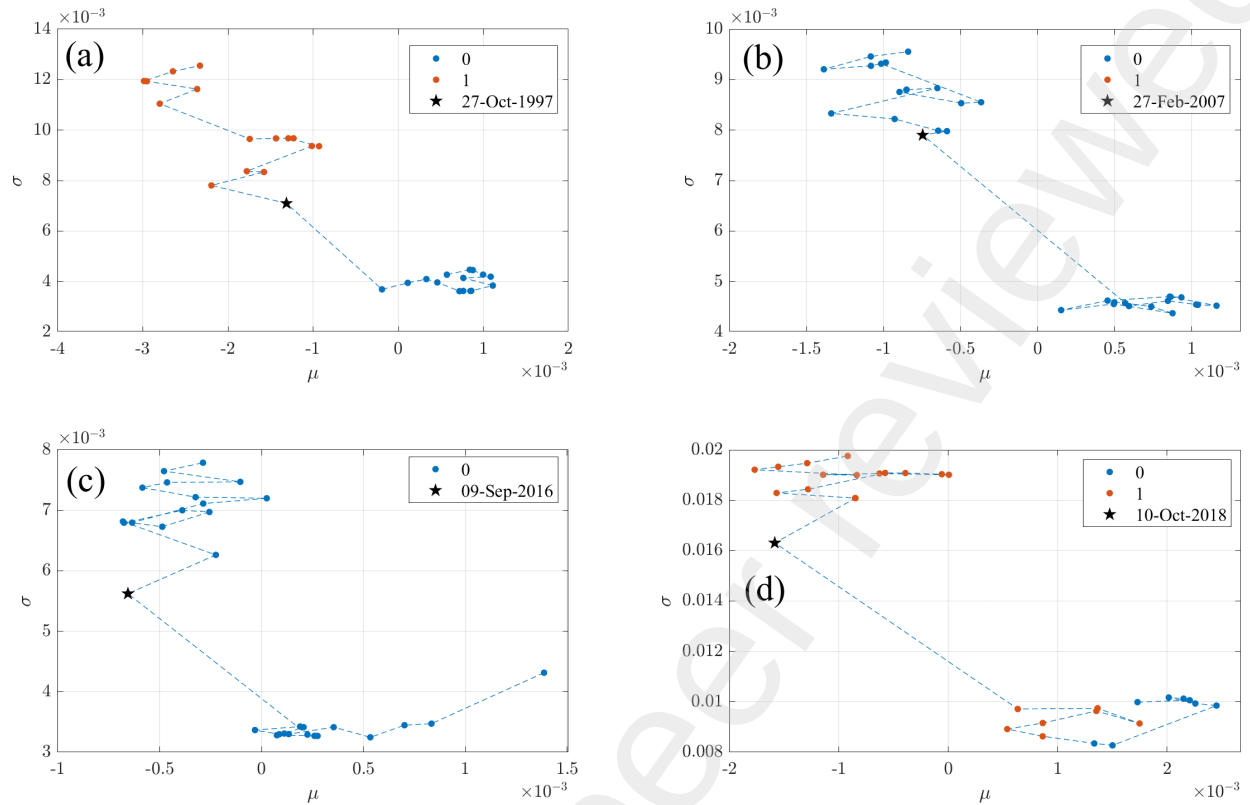


Fig. 12: Graphs represent statistical paths and clusters in the S&P 500 market before and after FID reaches its peak (black star) compared to HMM states during financial crises: (a) 1997, (b) 2007, (c) 2016, and (d) 2018.

deforms after the FID reaches its peak, suggesting that there is a regime switch reflected by the abrupt change in statistical moments. This change is mapped to a jump (star symbol) on the Euclidean coordinates (as a simplification of the actual coordinates on a curved statistical manifold) in Fig.12. It is clear that there are two groups (filled circles) of statistical parameters, before and after FID reaches its peak (denoted by the black star). In comparison, HMM is not intended to classify an abrupt change or a large jump in statistical parameters. Hence, HMM cannot effectively separate two distinct regimes on the statistical manifold in the short time period as particularly shown in Fig.12(b)-12(d) as sometimes the red and blue dots are mixed and sometimes only one type occurs (red and blue dots). As a result, FID has an advantage over HMM in the sense of measuring continuous regime changing as a transition over infinite states on a statistical manifold. Additionally, there is a big change in the return distribution at a high peak of FID. Although some of these peaks can occur outside of financial crises, they indicate the structural breaking of market behavior that can influences traders' investment strategies. This is indeed a situation where traders should consider whether their current strategy need adjustment.

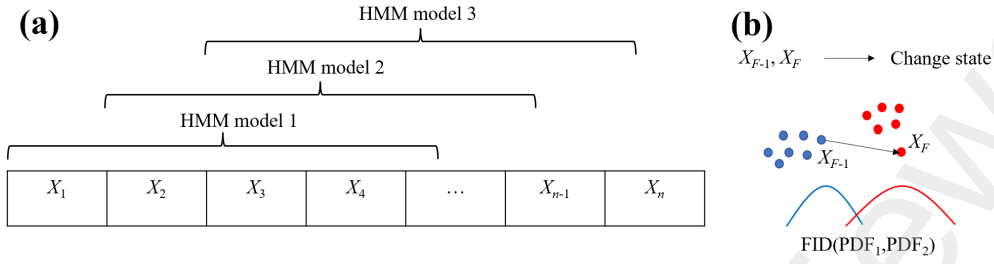


Fig. 13: The figure depicts the process of comparing regime changes: (a) moving windows of HMM and (b) determining whether the final state differs from the previous state.

3.4. Comparison of regime switching in HMM and FID

To compare the performance of HMM and FID in regime-switching detection, we utilize moving HMM to demonstrate the real-time detection, as illustrated in Fig.13a. We use half of the return data to fit HMM and continue moving the data window. We detect regime switching by considering whether the final state (X_F) of each moving HMM's identification differs from the previous state (X_{F-1}). If the state changes, we count it as a regime-switching signal. On the other hand, the same number of signals from FID are chosen from the top FID peaks. A high level of FID denotes a significant jump on the statistical manifold and refers to regime switching, as discussed in the previous section and illustrated in Fig.13b. Then, the efficacy of signals from both sources is determined by the difference in 30-day return distributions before and after each signal occurrence or the FID with lag 30. As shown in Fig.14, signals from FID can indicate a higher level of distribution change than those of HMM ($p\text{-value} \approx 1.4 \times 10^{-4}$). This means that FID outperforms HMM in identifying statistical changes or regime-switching when real-time data is used. Although the FID signal is measured based on daily change, it can be used as an early warning signal of regime switching in the near future. This is due to the fact that the FID distribution is a generalized extreme value. After a big jump, which is at the tail of the distribution, the size of jumps tends to be much smaller. As a result, the position of the return distribution on manifold is not far away from its previous position.

To summarize, FID and FIA could be useful for identifying the level of regime switching and the direction of change, as well as clustering on a statistical manifold, as demonstrated in the previous section. However, we should emphasize that this is a simple classification problem. Other applications of FIA and FID require further exploration and investigation, which we would like to leave for future research.

3.5. Results from STOXX600

The same analysis is performed on STOXX600, as shown in Appendix D. We found comparable results with those from S&P500. FID and FIA can ostensibly detect deformation in return distribution, indicating a financial market regime change. Furthermore, the Fisher metrics outperform HMM in clustering statistical parameters when there is a huge change in a short period of time.

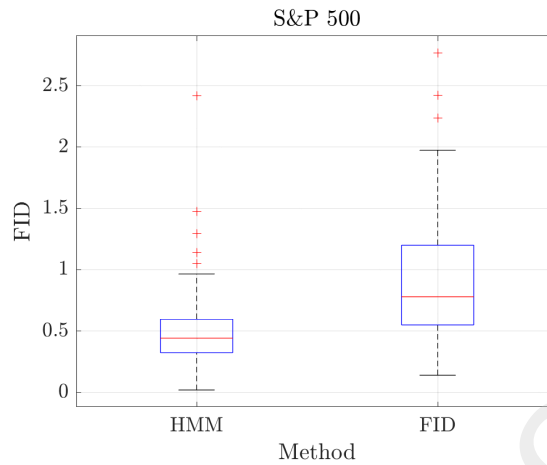


Fig. 14: The figure shows box plot of distribution deformation between 30 days before and after the signals from FID and HMM of S&P 500

4. Conclusion

In financial markets, regime switching is concern a major change in volatility and the distribution of returns. To discover regime switching occurrences, we employ information geometry as an instrument to capture changes in the return distribution. We discover that FID can be a powerful tool for analyzing changes in the return distribution and volatility since it can produce distinguishable peaks during periods of significant volatility changes or financial crises. FID not only reveals a distinct peak on the same date as volatility, but also represents extra information about changing the distribution that the volatility does not.

The FID distribution shows an opposing tendency to the stylized fact of a financial market, in which the cumulative distribution function of absolute return represents a shorter tail and approaches normal distribution as the lag time increases. In contrast, the FID distribution has a larger tail as the lag time increases. These findings show that the level of regime switching will change drastically when the lag time increases, which also suggests that other financial models associated with statistical parameters should be updated for their parameters to become more efficient. Furthermore, we discover that the distributions of the shuffled return data are significantly different from those of the actual data because of the nature of ordering in the return time series. The actual data has a longer tail whereas the shuffled data has a shorter tail as the lag time increases. We observe that the probability of obtaining the FIA in positive and negative values with an asymmetrical distribution increases with increasing lag time. Moreover, we noticed that when the lag time increases, the shift in mean and standard deviation indicates anticorrelation.

Finally, we compare the performance of FID and HMM approaches for detecting distribution deformation in the near future with real-time data. We discover that FID can indicate a high level of deformation better. This refers that FID is effective at detecting significant regime change in the sense of a shift in statistical parameters and distributions.

In summary, we can utilize the FID to detect regime switching in a financial market.

Furthermore, FID can reveal market behavior when HMM cannot since a limited state is defined, such as only high volatility and low volatility, whereas FID shows the level of regime switching instead. However, we remark that the size of considered window of the FID is crucial for investigation. This work contributes to understanding of market behaviors especially regime switching in terms of information geometry.

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Appendix A. Normality test, effect of time window, and robustness

To examine the validity of the normal distribution approximation, the influence of the time window, and the robustness of the FID, we apply our method to STOXX 600 (the stoxx Europe 600 index) from 2004 to 2022. This index contains a set number of 600 components covering major, medium, and small capitalization companies from 17 European countries.

We first examine the normality of return distribution with different sizes of time windows with the Anderson-Darling test [69]. This is a hypothesis testing with the null hypothesis being that the data is from a population with a normal distribution, and it is decided by the value of H . If $H = 0$, it indicates that the test fails to reject the null hypothesis. However, if $H = 1$, it indicates that the test rejects the null hypothesis. We test the normality for both S&P 500 and STOXX 600 indices as shown in Table A.1 and Table A.2, respectively. The results show that an increasing number of data in the return distribution, e.g. wider time windows, increases the percentage of time windows wherein data distribution are not normal.



Fig. A.1: Comparing window size effects with 30,60, and 120 trading days for S&P 500

The time window size also affects FID as shown in Fig.A.1 for S&P 500 and Fig.A.2 for STOXX 600. However, the significant peaks on the date of high volatility (R^2) are not generally affected by the window size. Moreover, there is a small delay of the FID peaks. This is believed to occur because of a large size of the time window containing more data on the distribution. Therefore, when computing consecutive return distribution, the FID gets



Fig. A.2: Comparing window size effects with 30, 60, and 120 trading days for STOXX 600

smaller as the two distribution are more similar to each other. On the contrary, a small size of time window contains a small number of data. If there is an extreme value added on the next time step, It will represent a big jump in FID.

Table A.1: Anderson-Darling test for normality of the return distribution of S&P 500 in different lengths of trading days N , where N_R is total number of return distributions tested.

N	N_R	$H = 0$ (frequency)	$H = 1$ (frequency)	$H = 0$ (%)	$H = 1$ (%)
10	6912	6500	412	94.04	5.96
30	6892	5947	945	86.29	13.71
60	6862	4956	1906	72.22	27.78
120	6802	3349	3453	49.24	50.76

Appendix B. Analysis of FID Peaks

We emphasize that the peaks of FID correspond to the distance of the return distribution compared to that one day earlier. Thus, an FID peak need not correspond to a crisis, but rather to a distinct deformation of the return distribution. Indeed, during a period of crisis or approaching a crisis, the change of distribution tends to occur due to a decrease in the average return and a surge of volatility. Thus, during a crisis, the FID peaks are easily observed and coincide with the peaks of R^2 .

Table A.2: Anderson-Darling test for normality of the return distribution of STOXX 600 in different lengths of trading days N , where N_R is total number of return distributions tested.

N	N_R	$H = 0$ (frequency)	$H = 1$ (frequency)	$H = 0$ (%)	$H = 1$ (%)
10	4563	4243	320	92.99	7.01
30	4543	3844	699	84.61	15.39
60	4513	3390	1123	75.12	24.88
120	4453	2081	2372	46.73	53.27

However, high FID peaks can occur during a relatively calm period when there is a significant deformation of the return distribution. To see this more analytically, we consider the equations (8) and (9), where FID for two Gaussian distributions is expressed by the function $D_F((\mu_1, \sigma_1), (\mu_2, \sigma_2))$. Since

$$F = \sqrt{((\mu_1 - \mu_2)^2 + 2(\sigma_1 - \sigma_2)^2)((\mu_1 - \mu_2)^2 + 2(\sigma_1 + \sigma_2)^2)} \geq 0,$$

and the logarithmic function is monotonically increasing, we obtain

$$\begin{aligned} D_F((\mu_1, \sigma_1), (\mu_2, \sigma_2)) &= \sqrt{2} \ln \left(\frac{F + (\mu_1 - \mu_2)^2 + 2(\sigma_1^2 + \sigma_2^2)}{4\sigma_1\sigma_2} \right) \\ &\geq \sqrt{2} \ln \left(\frac{2(\sigma_1^2 + \sigma_2^2)}{4\sigma_1\sigma_2} \right) := D_{F_0}(\sigma_1, \sigma_2) \end{aligned}$$

We call $D_{F_0}(\sigma_1, \sigma_2)$ the approximate FID, and it can be expressed as

$$D_{F_0}(\sigma_1, \sigma_2) = \sqrt{2} \ln \left[\frac{1}{2} \left(\frac{\sigma_1}{\sigma_2} + \frac{\sigma_2}{\sigma_1} \right) \right] \quad (\text{B.1})$$

Now one can see that $D_F((\mu_1, \sigma_1), (\mu_2, \sigma_2))$, hence FID, is bounded from below by $D_{F_0}(\sigma_1, \sigma_2)$, which can be large when either σ_1 or σ_2 is small, while the other is moderate. We recall that σ_1 and σ_2 represent the standard deviations of the return distributions from two consecutive days (with σ_1 being from an earlier day). Notably, the peaks during the calm periods 2012-2014 or 2016-2017 are still present, even though it is clear that the volatility R^2 is low in these periods; see also Fig.10(b).

Appendix C. Derivation of FIA

In this section, we derive the FIA formula, which is used to determine the direction of change in the distribution. We may utilize the Fisher information in the Cartesian coordinates to find the angle between two vectors given by

$$\cos \theta = \frac{\langle X, Y \rangle}{\|X\| \|Y\|}, \quad (\text{C.1})$$

where X is a vector that represents the change in μ and σ . This can be expressed as

$$X = \begin{pmatrix} \mu_2 - \mu_1 \\ \sigma_2 - \sigma_1 \end{pmatrix}, \quad (\text{C.2})$$

and $Y = (1, 0)^T$. Hence, this measures the angle between changing in statistical parameters (μ) and a unit vector that is parallel to the x axis. We recall that the Fisher information metric for a normal distribution is expressed as

$$g_{ij} = \begin{pmatrix} \frac{1}{\sigma_1^2} & 0 \\ 0 & \frac{2}{\sigma_1^2} \end{pmatrix}. \quad (\text{C.3})$$

Then,

$$\langle X, X \rangle = (\mu_2 - \mu_1 \quad \sigma_2 - \sigma_1) \begin{pmatrix} \frac{1}{\sigma_1^2} & 0 \\ 0 & \frac{2}{\sigma_1^2} \end{pmatrix} \begin{pmatrix} \mu_2 - \mu_1 \\ \sigma_2 - \sigma_1 \end{pmatrix} = \frac{(\mu_2 - \mu_1)^2}{\sigma_1^2} + \frac{2(\sigma_2 - \sigma_1)^2}{\sigma_1^2} \quad (\text{C.4})$$

Hence, the norm of vector X representing the length of the vector can be computed from

$$\|X\| = \sqrt{\langle X, X \rangle} = \frac{1}{\sigma_1} \sqrt{(\mu_2 - \mu_1)^2 + 2(\sigma_2 - \sigma_1)^2}. \quad (\text{C.5})$$

Similarly, we find the inner product $\langle Y, Y \rangle$:

$$\langle Y, Y \rangle = (1 \quad 0) \begin{pmatrix} \frac{1}{\sigma_1^2} & 0 \\ 0 & \frac{2}{\sigma_1^2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sigma_1^2}. \quad (\text{C.6})$$

This yields

$$\|Y\| = \frac{1}{\sigma_1}. \quad (\text{C.7})$$

Ultimately, we compute inner product $\langle X, Y \rangle$:

$$\langle X, Y \rangle = (\mu_2 - \mu_1 \quad \sigma_2 - \sigma_1) \begin{pmatrix} \frac{1}{\sigma_1^2} & 0 \\ 0 & \frac{2}{\sigma_1^2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{\mu_2 - \mu_1}{\sigma_1^2}. \quad (\text{C.8})$$

Putting all terms together into equation C.1, we obtain

$$\cos(\theta) = \frac{\mu_2 - \mu_1}{\sqrt{(\mu_2 - \mu_1)^2 + 2(\sigma_2 - \sigma_1)^2}}. \quad (\text{C.9})$$

Equivalently, the Fisher information angle can be written as follows:

$$\theta = \begin{cases} \cos^{-1} \left(\frac{\mu_2 - \mu_1}{\sqrt{(\mu_2 - \mu_1)^2 + 2(\sigma_2 - \sigma_1)^2}} \right), & \sigma_2 \geq \sigma_1, \\ -\cos^{-1} \left(\frac{\mu_2 - \mu_1}{\sqrt{(\mu_2 - \mu_1)^2 + 2(\sigma_2 - \sigma_1)^2}} \right), & \sigma_2 \leq \sigma_1. \end{cases} \quad (\text{C.10})$$

Appendix D. Data from STOXX600

Following the procedure as done for S&P 500, we first examined the statistical moments and overall information of STOXX600, as shown in D.1. We discovered that the mean μ and variance σ^2 values in unit of σ are quite close to each other with positive values. Skewness and kurtosis, on the other hand, exhibit the same pattern as those in S&P 500. When the time window is expanded, the skewness decreases and the kurtosis increases.

We discovered that the area of scatter also increases with lag time, including a stronger negative correlation and slope in linear regression. In contrast to the shuffled data, the tail of FID lengthens as the lag time increases, as shown in Fig.D.2-D.6.

We noticed that the likelihood of trend changes between $\Delta\mu$ and $\Delta\sigma$ indicates that both opposite trend change is approximately 50% similar that found in S&P 500, see Fig.D.7.

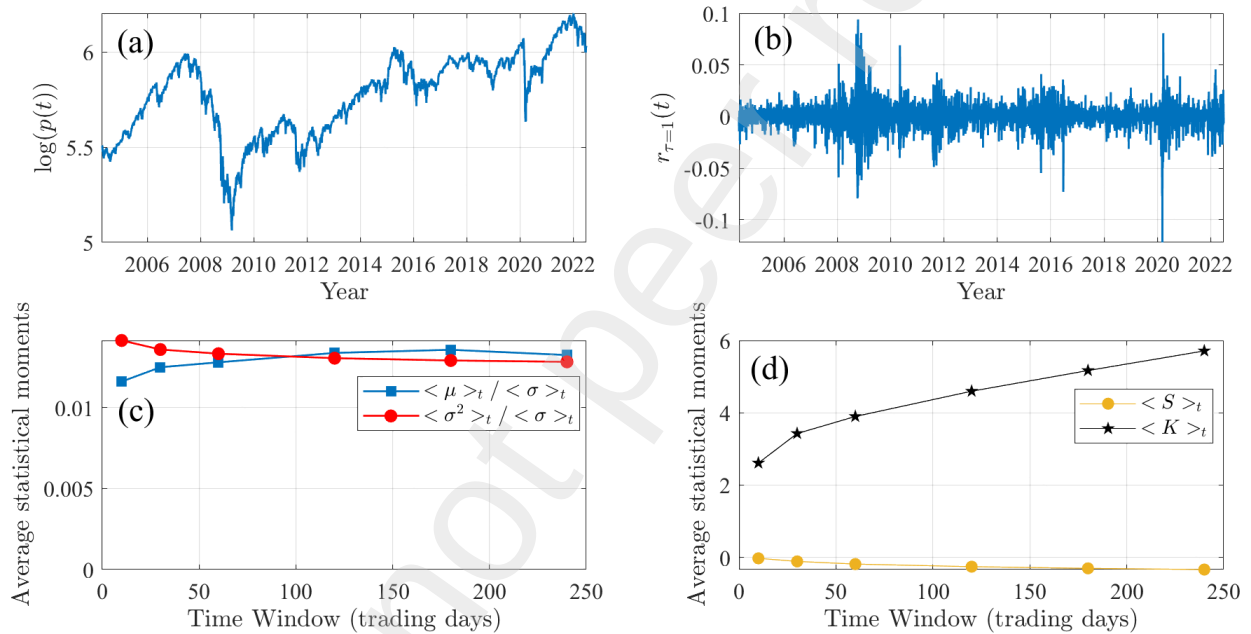


Fig. D.1: Financial data and statistical moments: (a) the logarithmic price of STOXX 600 from 2005 to 2022; (b) the daily logarithmic return of STOXX 600 from 2005 to 2022; (c) the average mean and variance in the standard deviation unit of the daily logarithmic return distributions with different sizes of time windows; (d) the average skewness ($\langle S \rangle_t$) and kurtosis ($\langle K \rangle_t$) of daily logarithmic return distribution with different sizes of time windows.

Moreover, we also observed changes in STOXX 600 during a financial crisis by using the Fisher information metrics as shown in Fig.D.8. We found that FID can still capture significant change in high volatility, and show distinct peaks following the level of change in the return distribution. It is found that normalized FID reaches its peak on September 19, 2008 but $\tilde{R}^2$ shows the highest value on October 13, 2008. The normalized FID is not necessarily at a high level when $\tilde{R}^2$ is. This is because normalized squared return and FID are different measurements that provide a different perspective. The return distributions and statistical moments on the Euclidean space for the top four highest FID are visualized in

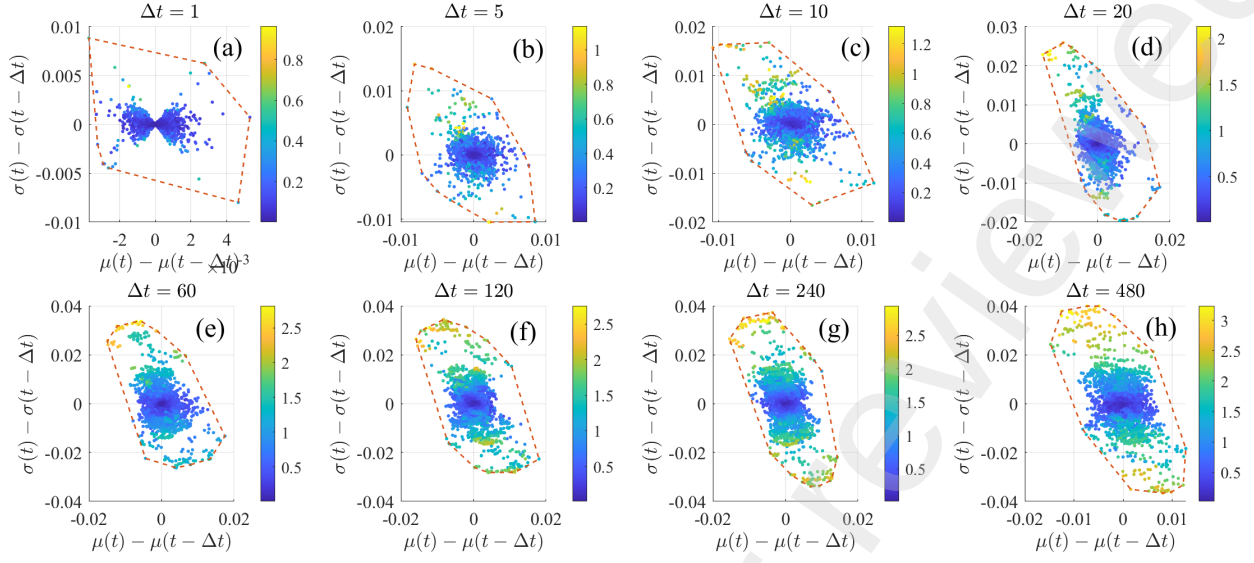


Fig. D.2: Scatter plots of STOX 600 of changes in μ and σ : (a) $\Delta t = 1$; (b) $\Delta t = 5$; (c) $\Delta t = 10$; (d) $\Delta t = 20$; (e) $\Delta t = 60$; (f) $\Delta t = 120$; (g) $\Delta t = 240$; (h) $\Delta t = 480$. The color of the scatters indicates the strength of FID, whereas the red dashed line denotes a convex hull region of scattering.

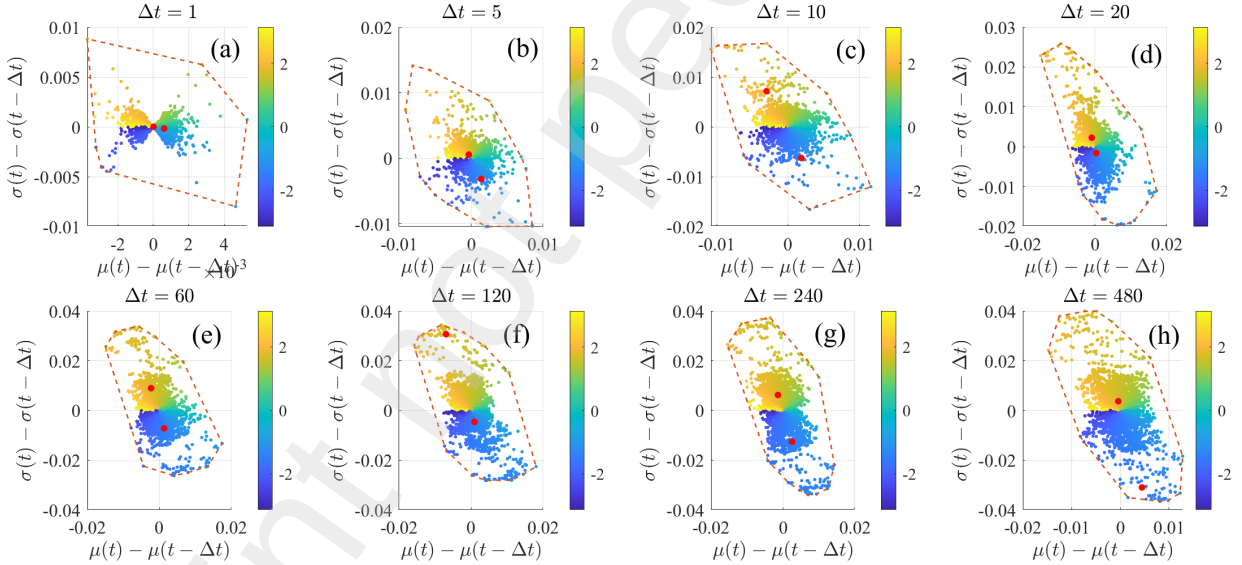


Fig. D.3: Scatter plot of STOX 600 of changes in μ and σ : (a) $\Delta t = 1$; (b) $\Delta t = 5$; (c) $\Delta t = 10$; (d) $\Delta t = 20$; (e) $\Delta t = 60$; (f) $\Delta t = 120$; (g) $\Delta t = 240$; (h) $\Delta t = 480$. The color of the scatter indicates the strength of FIA, whereas the red dashed line denotes a convex hull region of scattering. The red dots depict the regions where positive and negative Fisher information angles are most likely to be found.

Fig.D.10 and Fig.D.11, respectively. This demonstrates that, unlike HMM, FID can cluster statistical parameters in the sense of regime switching with infinite states.

Additionally, with the same measure as in Section 3.4, we found that the deformation detection by FID is still better than HMM, albeit with less significance (p -value $\approx 6.9 \times 10^{-2}$).

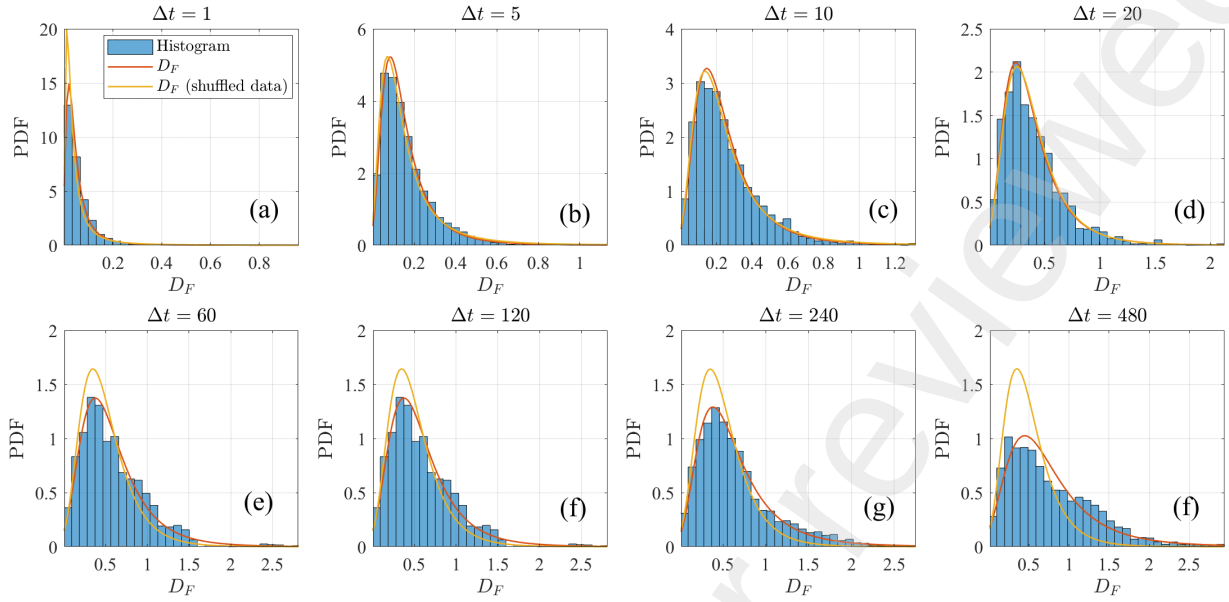


Fig. D.4: The probability density functions (PDF) of FID from original and shuffled data of returns of STOXX 600: (a) $\Delta t = 1$; (b) $\Delta t = 5$; (c) $\Delta t = 10$; (d) $\Delta t = 20$; (e) $\Delta t = 60$; (f) $\Delta t = 120$; (g) $\Delta t = 240$; (h) $\Delta t = 480$ days. Red lines are the fitting of real data generalized extreme value distributions, whereas the yellow lines depict the average of the FID distributions from the shuffled data 1000 configurations.

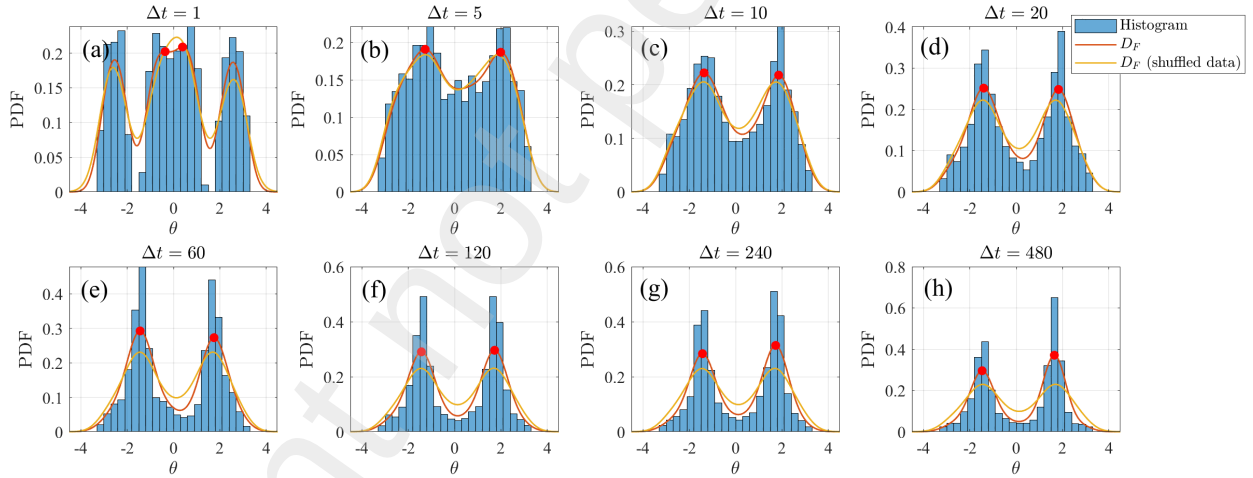


Fig. D.5: The probability density function (PDF) of FIA from original and shuffled data of returns of STOXX 600: (a) $\Delta t = 1$; (b) $\Delta t = 5$; (c) $\Delta t = 10$; (d) $\Delta t = 20$; (e) $\Delta t = 60$; (f) $\Delta t = 120$; (g) $\Delta t = 240$; (h) $\Delta t = 480$ days. Red lines are the fitting of real data Kernel distributions, whereas the yellow lines depict the average of the Fisher information angle distributions from shuffled data 1000 configurations.

655 Fig.D.9 also provides a boxplot that represents these two groups.

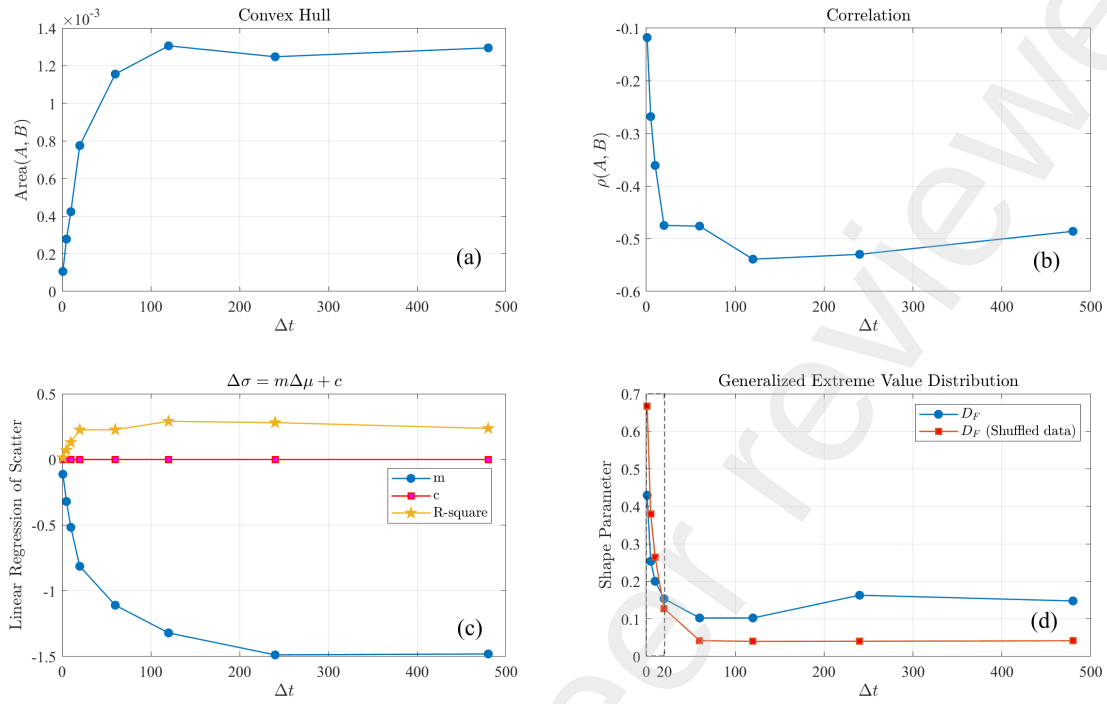


Fig. D.6: Statistical features of change in mean μ and standard deviation σ of STOXX 600: (a) the convex hull area of scattering region; (b) correlation between changes in μ and σ ; (c) linear regression of scatter; (d) comparing tail index between real data and shuffled data with the generalized extreme value distribution.

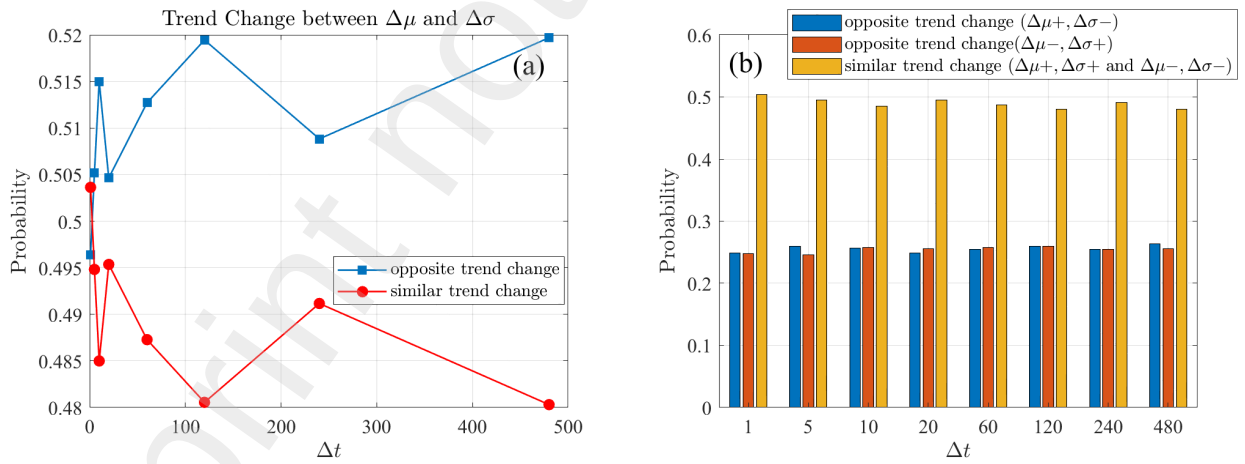


Fig. D.7: Comparison of trend changes of STOXX 600 between $\Delta\mu$ and $\Delta\sigma$ as the lag time Δt increases: (a) the probability of trend changes between $\Delta\mu$ and $\Delta\sigma$; (b) decomposition of the probability of opposite trend changes between $\Delta\mu$ and $\Delta\sigma$. The plus symbol represents an increased value in the signal; the minus symbol a decreased value. In (b), the dark blue and orange bars indicate opposite trend changes between signals, and the yellow ones a similar trend change.

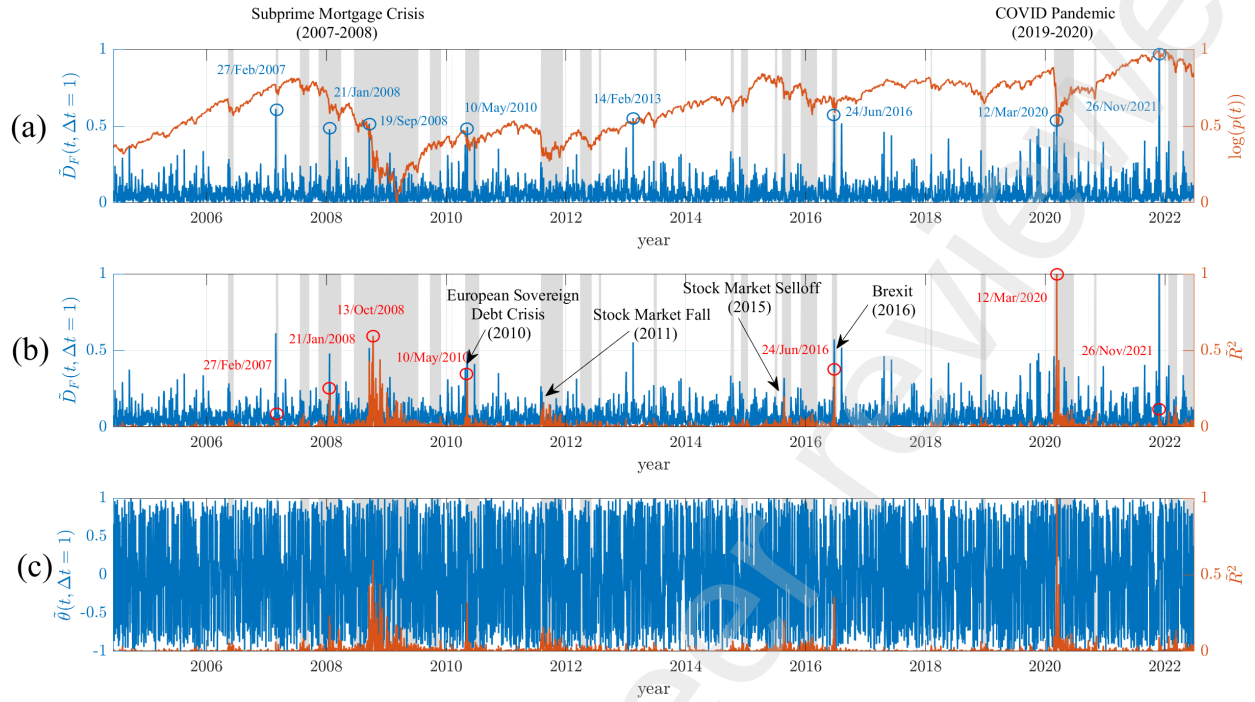


Fig. D.8: Superposed graphs of normalized FID and financial quantities in different financial periods of STOXX 600: (a) normalized FID versus normalized logarithmic price; (b) normalized FID versus normalized volatility $\tilde{R}^2$; (c) normalized FIA versus $\tilde{R}^2$. The gray-shaded areas illustrate the high volatility periods defined by HMM; the black arrows indicate short periods of high volatility. The blue circles represent notable peaks of FID which occurred on specified dates.

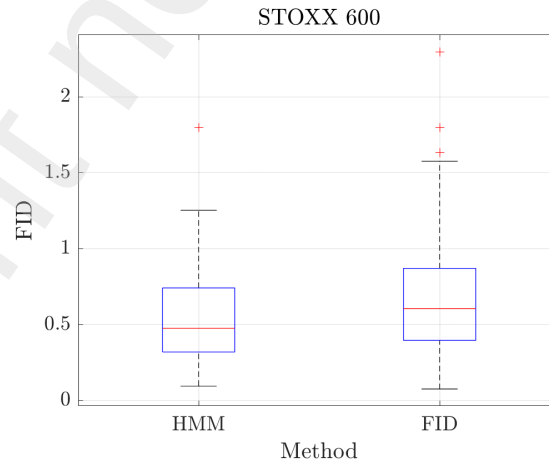


Fig. D.9: Box plots of distribution deformation between 30 days before and after the signals from FID and HMM of STOXX 600.

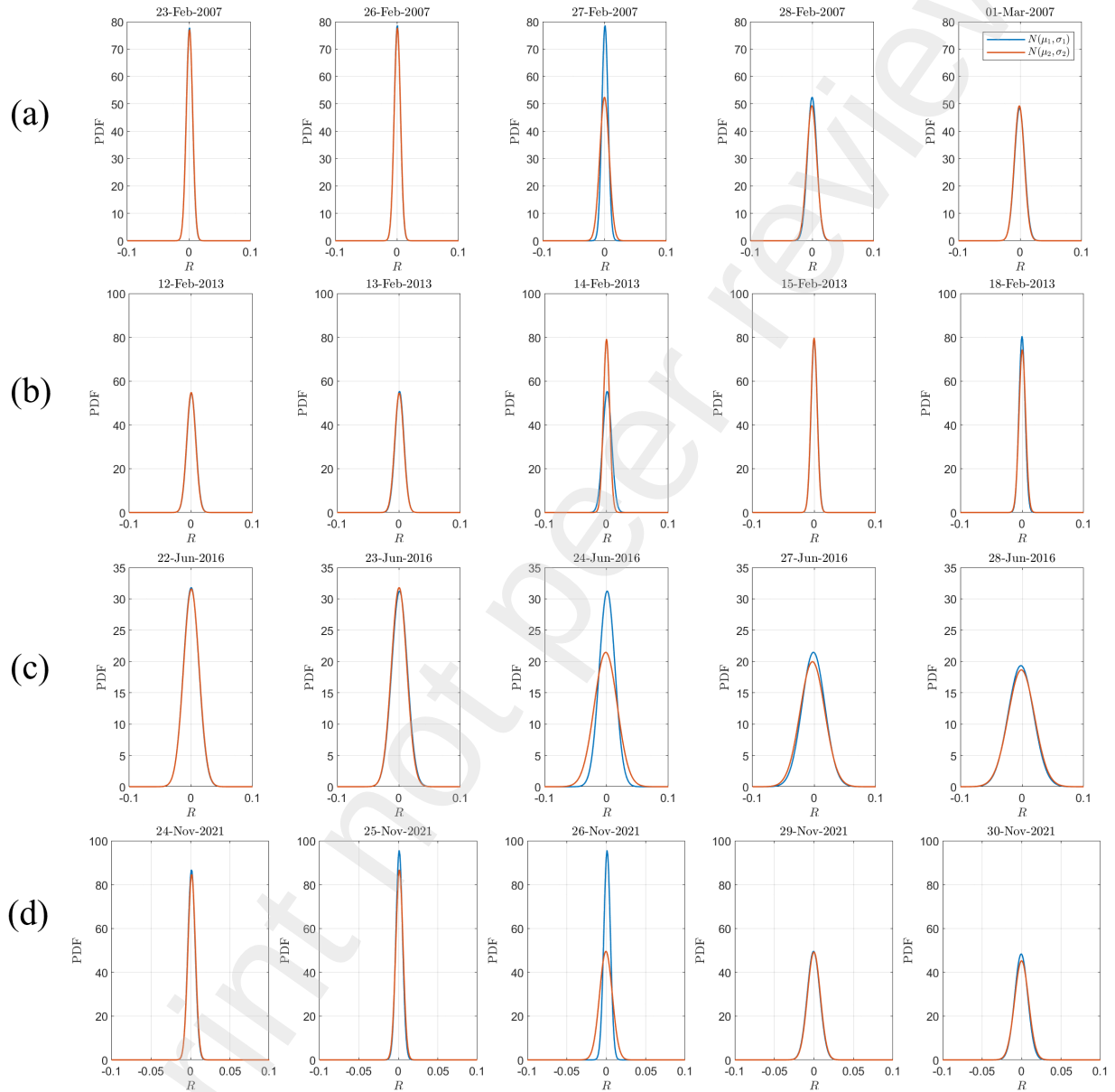


Fig. D.10: Graphs exhibit the return distributions in the STOXX 600 market that correspond to FID peaks in several periods: (a) 2007, (b) 2013, (c) 2016, and (d) 2021. Top four highest FID peaks are chosen.

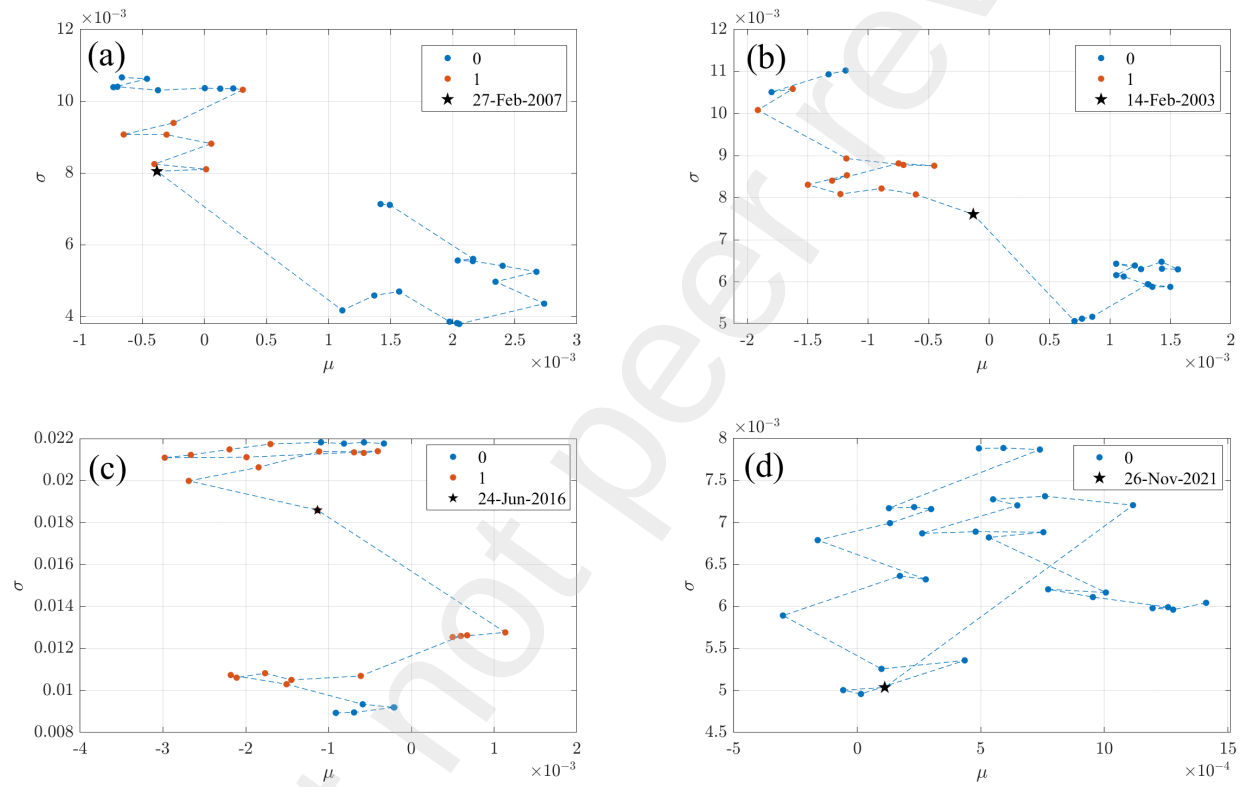


Fig. D.11: Graphs represent statistical paths in the STOXX 600 market before (blue circles) and after (red circles) FID reaches its peak: (a) 2007, (b) 2013, (c) 2016, and (d) 2021.